

Bonus Homework 6: Quantum Amplitude Estimation

Quantum Computing Training 2026

This bonus homework explores applications of quantum amplitude estimation (QAE). In each problem, your goal is to express a useful quantity as a probability

$$a = \Pr[\text{good outcome}],$$

and then estimate that probability using QAE.

You may implement QAE using the lecture-style construction: a state-preparation circuit A , a marked-state phase flip S_{marked} , the corresponding Grover operator

$$Q = -AS_0A^\dagger S_{\text{marked}},$$

and phase estimation on Q . When QAE returns an integer outcome y using m evaluation qubits, use the identity

$$\hat{a} = \sin^2\left(\pi \frac{y}{2^m}\right).$$

Because QPE may return either phase direction, the outcomes y and $2^m - y$ can encode the same amplitude. When you see two symmetric peaks, convert both using the formula above.

Problem 1: Quantum Counting with Hidden Marked States

In Grover's algorithm, the optimal number of Grover iterations depends on the number of marked states M . If M is unknown, QAE can estimate it. This is called *quantum counting*.

Consider a search space of size

$$N = 16,$$

represented using a 4-qubit register. There is a Boolean oracle for an unknown function

$$f : \{0, 1\}^4 \rightarrow \{0, 1\}.$$

The marked states are those with $f(x) = 1$.

For this problem, use the following hidden marked set inside your oracle:

$$\mathcal{M} = \{|0110\rangle, |1011\rangle, |1110\rangle\}.$$

Pretend that QAE does not know this set in advance. The set is provided only so that you can build and debug the oracle.

1.a. Prepare the uniform superposition

$$A|0000\rangle = H^{\otimes 4}|0000\rangle = \frac{1}{4} \sum_{x=0}^{15} |x\rangle.$$

What is the marked fraction

$$a = \Pr[f(x) = 1]$$

for the marked set above?

1.b. Build a phase oracle S_f that applies

$$S_f |x\rangle = (-1)^{f(x)} |x\rangle.$$

In Qiskit, you may implement this by converting each marked bit string into $|1111\rangle$ using X gates, applying a 4-controlled phase flip, and then undoing the X gates.

1.c. Build the QAE Grover operator

$$Q = -AS_0A^\dagger S_f,$$

where S_0 flips the phase of $|0000\rangle$.

1.d. Run QAE using

$$m = 3, 4, 5$$

evaluation qubits. For each m , record the largest few measurement probabilities over the evaluation register.

1.e. For the most likely nonzero QAE outcome y , compute

$$\hat{a} = \sin^2\left(\pi \frac{y}{2^m}\right), \quad \widehat{M} = 16\hat{a}.$$

Compare $\widehat{M}$ with the true value of M .

1.f. Use your estimate to choose a Grover iteration count

$$\hat{r} = \left\lfloor \frac{\pi}{4} \sqrt{\frac{16}{\widehat{M}}} \right\rfloor.$$

Run ordinary Grover search with $\hat{r}$ iterations. What is the probability of measuring a marked state?

Problem 2: Integral Estimation via QAE

In this problem, you will estimate a simple integral by converting it into an amplitude-estimation problem.

Estimate

$$I = \int_0^1 x^2 dx.$$

The exact value is

$$I = \frac{1}{3}.$$

Approximate the integral using $N = 8$ midpoint samples:

$$x_j = \frac{j + 1/2}{8}, \quad j = 0, 1, \dots, 7.$$

The midpoint estimate is

$$I_8 = \frac{1}{8} \sum_{j=0}^7 x_j^2.$$

2.a. Compute the eight values x_j and x_j^2 . Compute I_8 classically and compare it with $1/3$.

2.b. Use a 3-qubit input register to prepare

$$\frac{1}{\sqrt{8}} \sum_{j=0}^7 |j\rangle.$$

Add one objective qubit initialized to $|0\rangle$.

2.c. For each basis state $|j\rangle$, apply a controlled rotation to the objective qubit so that

$$|j\rangle |0\rangle \mapsto |j\rangle \left(\sqrt{1-x_j^2} |0\rangle + \sqrt{x_j^2} |1\rangle \right).$$

Use a controlled $R_y(\theta_j)$ rotation with

$$\sin^2\left(\frac{\theta_j}{2}\right) = x_j^2.$$

What is θ_j in terms of x_j ?

2.d. Show that the probability of measuring the objective qubit in state $|1\rangle$ is

$$a = \frac{1}{8} \sum_{j=0}^7 x_j^2 = I_8.$$

Verify this probability using a statevector simulator before running QAE.

2.e. Use QAE to estimate a . Try

$$m = 3, 4, 5$$

evaluation qubits. Make a table with columns

$$m, \quad y_{\max}, \quad \hat{a}, \quad |\hat{a} - I_8|, \quad |\hat{a} - 1/3|.$$

2.f. Repeat with $N = 16$ midpoint samples using a 4-qubit input register. Which error is now larger: the discretization error from midpoint sampling, or the QAE estimation error?

Problem 3: Estimating the Probability of Network Failure

Often, people are interested in calculating the probability of a rare bad event. QAE can estimate this probability if the bad event is encoded as the marked subspace.

Consider a small network with three links. Link 0 is a direct path from source to target. Links 1 and 2 form a backup two-hop path. The network succeeds if either

link 0 works

or

links 1 and 2 both work.

Equivalently, the network fails if

link 0 fails and at least one of links 1 or 2 fails.

Let the independent link failure probabilities be

$$p_0 = 0.10, \quad p_1 = 0.20, \quad p_2 = 0.15.$$

Use one qubit per link, where

$$|1\rangle = \text{failed}, \quad |0\rangle = \text{working}.$$

3.a. Classically compute the exact network failure probability:

$$p_{\text{fail}} = p_0 (1 - (1 - p_1)(1 - p_2)).$$

3.b. Prepare the 3-qubit probability distribution over link states. For each link i , prepare

$$\sqrt{1 - p_i} |0\rangle + \sqrt{p_i} |1\rangle.$$

What R_y angle should be used on link qubit i ?

3.c. Add one objective qubit. Construct a reversible predicate that flips the objective qubit if and only if the network has failed:

$$x_0 = 1 \quad \text{and} \quad (x_1 = 1 \text{ or } x_2 = 1).$$

You may use CNOT gates, Toffoli gates, X gates, and temporary ancillas.

3.d. Verify your predicate with a truth table. It should mark exactly

$$101, \quad 110, \quad 111,$$

where the bit string is $x_0x_1x_2$.

3.e. Treat the objective qubit state $|1\rangle$ as the good state. Use QAE to estimate the failure probability. Try

$$m = 3, 4, 5$$

evaluation qubits. Compare the QAE estimates with the exact value from part (a).

3.f. Compare QAE against direct sampling. Simulate measuring the prepared link-state circuit with

$$S = 100, 1000, 10000$$

shots and estimate the failure probability by the fraction of failed samples. How does the sampling error compare with your QAE estimates?

3.g. Change the failure probabilities to

$$p_0 = 0.02, \quad p_1 = 0.10, \quad p_2 = 0.10.$$

Repeat the experiment. What becomes harder when the marked event is rare?

Problem 4: A Toy Model for Option Pricing

Quantum amplitude estimation is often discussed as a possible tool for Monte Carlo-style finance problems. In this problem, you will estimate the expected payoff of a European call option in a small discrete model.

A European call option with strike price K has payoff

$$g(S) = \max(S - K, 0),$$

where S is the asset price at maturity.

Consider a toy model where the final asset price S takes four possible values:

$$S_0 = 80, \quad S_1 = 100, \quad S_2 = 120, \quad S_3 = 140.$$

The probabilities are

$$q_0 = 0.10, \quad q_1 = 0.40, \quad q_2 = 0.30, \quad q_3 = 0.20.$$

Let the strike price be

$$K = 105,$$

and let the payoff scale be

$$P_{\max} = 40.$$

4.a. Compute the payoff for each price:

$$g_j = \max(S_j - K, 0).$$

Then compute the exact expected payoff

$$\mathbb{E}[g(S)] = \sum_{j=0}^3 q_j g_j.$$

4.b. If the annual risk-free rate is

$$r = 0.05$$

and the maturity is

$$T = 1,$$

compute the discounted option value

$$V = e^{-rT} \mathbb{E}[g(S)].$$

4.c. Use two qubits to represent the four price outcomes. Prepare the probability distribution

$$\sum_{j=0}^3 \sqrt{q_j} |j\rangle.$$

You may use Qiskit's state-preparation tools or construct the state manually.

4.d. Add one objective qubit. For each price state $|j\rangle$, apply a controlled rotation so that

$$|j\rangle |0\rangle \mapsto |j\rangle \left(\sqrt{1 - g_j/P_{\max}} |0\rangle + \sqrt{g_j/P_{\max}} |1\rangle \right).$$

What rotation angle should be used for each j ?

4.e. Show that the probability of measuring the objective qubit in state $|1\rangle$ is

$$a = \frac{\mathbb{E}[g(S)]}{P_{\max}}.$$

Verify this probability with a statevector simulator before running QAE.

4.f. Use QAE to estimate a , then estimate the expected payoff using

$$\widehat{\mathbb{E}[g(S)]} = P_{\max} \hat{a}.$$

Finally estimate the option value

$$\hat{V} = e^{-rT} P_{\max} \hat{a}.$$

4.g. Run QAE with

$$m = 2, 3, 4, 5$$

evaluation qubits. Make a table comparing

$$m, \quad \hat{a}, \quad \widehat{\mathbb{E}[g(S)]}, \quad \hat{V}$$

against the exact values.

4.h. Discuss one limitation of this toy model. For example: the asset distribution is tiny, the payoff is discretized, and the state-preparation circuit is assumed rather than derived from market data.